

ANH-DUNG HUYNH

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Nha

Trang, Vietnam (Relocating to Ho Chi Minh City)

SUMMARY

AI Engineer specializing in NLP, TTS, LLM applications (LangChain, LangGraph), and high-reliability backend systems (Python, Java, modern databases, message queues). Experienced in developing and deploying production AI-applied products, vector search pipelines, and microservices.

EDUCATION

UNIVERSITY OF INFORMATION TECHNOLOGY, VNU-HCM | *B.S. in Computer Science* Sept 2022 – Sept 2025

- 2nd Place - UIT Data Science Challenge 2024

EXPERIENCE

FPT SOFTWARE

Vietnam

AI Engineer

Sept 2025 – Present

- Led a 5-person AI development team, collaborating with PM, PO, and external vendors across iterative delivery cycles to deploy enterprise AI automation systems.
- Took end-to-end technical ownership of a production conversational workflow system automating internal workflows and structured ticket orchestration from architecture through cloud deployment.
- Architected a multi-layer query routing engine combining LLM reasoning, semantic vector similarity, and deterministic rule matching, drastically curtailing unnecessary model calls and cloud inference costs.
- Engineered an asynchronous, message-driven backend backbone with RabbitMQ, eliminating synchronous API bottlenecks and scaling concurrent throughput under load.
- Solo-designed an agentic orchestration platform in Java (Spring Boot) featuring a declarative dynamic tool-dependency graph that calculates shortest execution paths based on parameter availability, context history, and vector retrieval (Qdrant).
- Enforced enterprise-grade reliability using Pydantic validation, structured exception hierarchies, distributed tracing, and centralized logging across Python and Java microservices.

HPT CORPORATION

Ho Chi Minh City, Vietnam

AI Engineer Intern

Jan 2025 – Jun 2025

- Conducted research and development for Legal NLP, designing high-precision RAG pipelines and domain-specific Named Entity Recognition (NER).
- Trained domain-adapted NER models using PyTorch and Hugging Face transformers, utilizing synthetic data augmentation to resolve severe class imbalance.
- Designed semantic chunking and intelligent document segmentation strategies to optimize retrieval relevance across complex legal corpora.
- Conducted controlled ablation experiments comparing embedding models and architectures, benchmarking precision, recall, F1, and inference latency for R&D leadership.

PROJECTS

DYNAMIC TOOL-ORCHESTRATION & GRAPH EXECUTION PLATFORM

2025 – Present

- Technologies: Java (Spring Boot), Python, Qdrant, Kafka, PostgreSQL, Redis
- Solo-architected an enterprise agent orchestration engine with an admin portal enabling dynamic tool declaration, typed parameter schemas, and input-output aliasing.
- Modeled tool dependencies as a directed network, computing optimal shortest paths (minimizing missing parameters) integrated with conversation history, user profiles, and vector retrieval.
- Built a high-throughput event-driven microservices architecture using Spring Boot, Kafka, and Qdrant, accompanied by a lightweight Python service for asynchronous knowledge ingestion.

PRODUCTION CONVERSATIONAL WORKFLOW & TICKET ORCHESTRATOR

2025 – Present

- Technologies: Python, LangChain, LangGraph, RabbitMQ, Milvus, Redis, Docker
- Led technical development for a production AI conversational engine automating enterprise workflows and internal support ticket routing.
- Designed a multi-tier routing pipeline combining deterministic matching, semantic vector search, and LLM reasoning to optimize response latency and token costs.
- Decoupled system components with a message-driven RabbitMQ bus, ensuring resilient, non-blocking execution under heavy concurrency.

AI INSURANCE AGENT MENTORING PLATFORM

2025 – Present

- Technologies: Python, LangGraph, FastAPI, Vector DB
- Developed an AI-powered simulation and mentoring platform for insurance agents providing contextual roleplay evaluation and automated feedback loops.

MULTI-MODAL SARCASM DETECTION & LLM FEATURE ABLATION | [Link](#)

2024

- Technologies: Python, PyTorch, ViSoBERT, CafeBERT, Hugging Face
- Researched and implemented multi-modal sarcasm detection on social media data for the UIT Data Science Challenge 2024 (2nd Place).
- Conducted extensive ablation on intermediate attention layers from bidirectional encoders and autoregressive models, evaluating max/mean/min pooling representations against shallow feed-forward heads.

SKILLS

AI & LLM Frameworks LangChain, LangGraph, NLP, TTS, Pydantic, Vector Search, PyTorch, Hugging Face, Transformers, Agentic Orchestration

Programming Languages Python, Java, Rust, SQL, FastAPI, Spring Boot

Databases & Message Queues Milvus, Qdrant, PostgreSQL, MySQL, MongoDB, SQLite, Redis, Kafka, RabbitMQ

Infrastructure & DevOps Docker, Docker Compose, Azure, VPS, GitLab CI/CD, GitHub Actions

Languages Vietnamese (Native), English (Proficient (TOEIC 980 LR, 320 SW)), French (Independent (DELF B2))

CERTIFICATIONS

AI Anomaly Detection, NVIDIA

A W A R D S

2024 2nd Place - UIT Data Science Challenge 2024, Solved multi-modal sarcasm detection on social media by conducting comprehensive ablation studies on LLM latent features (ViSoBERT, CafeBERT, GPT families) paired with custom pooling strategies and classification heads.